Accounting for Cryptocurrency Value

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Abstract

This paper examines the role of the information disclosed on blockchains in the cryptocurrency market. We find that blockchain disclosure on user adoption, measured as the number of new addresses, is highly value-relevant in the cryptocurrency market. Surprises in the disclosed number of new addresses explain 8% of the variation in cryptocurrency returns. The disclosed information is more value-relevant if the quality of the disclosure is higher and the cryptocurrency is larger in size. Unlike traditional markets, we do not find pre- or post-drift around the disclosure of new address information. The presence of strong market reactions at the disclosure and the absence of drifts around it highlight the distinct features of the information environment in this market, and provide a benchmark case where information is disclosed publicly and (almost) continuously. Lastly, we construct the price-to-new address ratios and find that they negatively predict future returns–a cryptocurrency value effect.

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I Introduction

The cryptocurrency market has gained widespread attention as its total market capitalization surpasses the trillion dollar mark. While providing a new way of financing for ventures and a new investment vehicle for investors, how to properly account for the intrinsic value of cryptocurrency is a matter of extensive debate among investors and regulators. The Financial Accounting Standards Board (FASB) has recently recommended fair value measurement for cryptocurrency assets, which is based on their market prices.[1] However, critics argue that the cryptocurrency market is highly speculative and volatile, making market prices unreliable and misleading indicators of their intrinsic value. In light of the debate, it is essential to understand alternative methods of valuing cryptocurrencies that go beyond market-based information.

The challenge lies in the fact that cryptocurrencies do not have a standardized accounting framework, unlike companies that have regulated accounting standards for financial measurements. Prior research highlights the role of voluntary disclosures, such as white paper, in the cryptocurrency market (e.g., Bourveau et al. (2022)). However, these voluntary disclosures are concentrated during the initial coin offering (ICO) stage and are often considered unreliable.[2] On the other hand, cryptocurrencies are the first asset class that discloses a complete history of economic activities recorded on the blockchain. As pointed out by the Institute of Chartered Accountants in England and Wales (ICAEW), the blockchain can be viewed as an accounting system that accurately and truthfully records and discloses the economic activities of cryptocurrencies.[3] Such disclosures are publicly and (almost) continuously available, and thus potentially provide a source of reliable information for valuing cryptocurrencies.

In this paper, we attempt to explore the role of the accounting information disclosed on blockchains

[1] See, e.g., https://www.wsj.com/articles/standards-setter-floats-new-rule-on-crypto-accounting-11675267296.

[2] The Securities and Exchange Commission (SEC) warned about fraudulent information contained in these voluntary disclosures (e.g., https://www.sec.gov/news/press-release/2018-88). The industry also identifies the so-called "white paper plagiarism scam," where scammers copy the white paper of a promising ICO and launch it using a similar or different name to deceive investors (e.g., https://cointelegraph.com/news/did-you-fall-for-it-13-ico-scams-that-fooled-thousands).

[3] The description from ICAEW about blockchain can be found in ICAEW (2018). The idea of using blockchain activities as a source of "accounting information" is also discussed in Amiram et al. (2022).

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(on-chain information) in valuing cryptocurrencies. What type of on-chain information may be relevant to cryptocurrency valuation? Recent models of dynamic cryptocurrency pricing highlight that cryptocurrencies have a strong network effect and that their value should increase when more users join the network, known as user adoption (e.g., Sockin and Xiong (2020) and Cong et al. (2021a)). Building on this theoretical literature, we propose and comprehensively analyze a measure of user adoption disclosed on blockchains to value cryptocurrencies–the number of new addresses.[4] Empirically, using the value-relevance tests developed in financial accounting research that examine the information content of earnings in the equity market (e.g., Lev and Zarowin (1999) and Srivastava (2014)), we find that information disclosed on new addresses is highly value-relevant for cryptocurrencies.[5] Furthermore, we explore the unique features of blockchain disclosure to revisit several classic questions for which the amount and revelation of information are central.

We start by examining the usefulness and relevance of the information disclosed on new addresses for the cryptocurrency market. We apply the value-relevance method and investigate the market reaction to the incremental information, or surprises, embedded in the disclosed number of new addresses. Our results indicate that this incremental information is highly value-relevant. On average, the surprises in the disclosed number of new addresses can account for more than 8% of the variations in cryptocurrency returns in the cross-section, which is comparable to that of earnings in the equity market.[6] Importantly, the explanatory power of new addresses remains similar controlling for changes in transaction volume and Google searches, which are proxies for changes in investor demand for and attention to cryptocurrencies, respectively. Moreover, we show that many economic determinants of the value-relevance of earnings in the equity market (see, e.g.,

[4]In this paper, we focus on *new* user adoptions by examining new addresses, as opposed to the number of active addresses. Although user exit should also affect the network size in principle, in practice, there is no closure of addresses, making it difficult to measure user exit. Measures of active addresses, thus, inevitably involve assumptions on the time periods the addresses are active. Our measure involves little discretion and the results can be thought of as a lower bound of the value-relevance of the user adoption effect in the cryptocurrency market.

[5]Throughout the paper, we draw an analogy between the number of new addresses and earnings because we use the value-relevance tests of earnings to motivate our research design. It is important to note, however, that while we draw this analogy, we do not consider the number of new addresses to be equivalent to earnings.

[6]The equity literature suggests that earnings surprises on average are able to explain about 5% of the variations in stock returns around the earnings announcements (e.g., Lev (1989); Lev and Gu (2016)).

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Kothari (2001) and Schipper and Vincent (2003)), such as quality and size, have counterparts in the cryptocurrency market, and that these counterparts are also important in determining the value-relevance of new addresses in the cryptocurrency market. Specifically, we find that the value-relevance of new addresses is higher when the growth rate of new address is more persistent or less volatile, consistent with the idea that the value-relevance of accounting numbers increases in the quality of the information. We also find that the value-relevance of new addresses is higher for larger cryptocurrencies, consistent with predictions from cryptocurrency valuation theories that highlight the network effect in this market.

One important difference between cryptocurrencies and traditional markets, such as equities, lies in how accounting information about underlying economic activities is disclosed. For cryptocurrencies, such information is made public and nearly continuous on the blockchain.[7] In contrast, information on firm activities is only disclosed periodically. We draw on this distinction and revisit several classic questions based on evidence from the equity market to understand the information environment in the cryptocurrency market.

First, we test whether the cryptocurrency market's reaction to the disclosure of new accounting information is sluggish, as is observed in the equity market. Research has shown that the stock market's response to earnings surprises can predict future returns, a phenomenon known as post-announcement drifts (e.g., Ball and Brown (1968); Bernard and Thomas (1989)). Among many alternative explanations, a common view is that the equity market is slow in incorporating earnings information. However, in the cryptocurrency market, where the information disclosed on new addresses is transparent and made available in a timely manner, it is ex ante unclear whether this phenomenon will be observed. On the one hand, return drifts following information disclosure may occur because attention is a scarce resource (e.g., Peng and Xiong (2006)), and continuous revelation of information can lead to strong under-reaction from investors. On the other hand, more frequent disclosure of accounting information may allow prices to reflect the information more quickly (e.g., Butler et al. (2007)). We find that, unlike the equity market, surprises in the

[7] For example, Bitcoin discloses information about every ten minutes.

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disclosed number of new addresses do not forecast cryptocurrency returns in the future, consistent with the view that transparent and timely information disclosure facilitates information acquisition.

Second, we examine the potential presence of return drifts prior to the disclosure of new address information. In the equity market, research finds that stock returns tend to precede important informational events such as earnings, a phenomenon known as pre-announcement drifts (e.g., Ball and Brown (1968)). The literature commonly attributes the phenomenon to the discrete nature of disclosure and the occurrence of information leakages between the discrete disclosures. In contrast, we find that cryptocurrency returns do not precede disclosures of new address information in the pre-disclosure period, consistent with the view that more frequent public disclosure reduces potential information leakages. The results also suggest that the value-relevance of information disclosed on new addresses is unlikely to be explained by user adoption in response to changes in cryptocurrency prices. Overall, the findings highlight the distinct features of the information environment in the cryptocurrency market.

Lastly, to further explore the role of accounting information in the cryptocurrency market, we use the number of new addresses to construct a financial ratio for cryptocurrencies: the price-to-new address ratio (*pa* ratio). We find that the *pa* ratio captures the cross-section of expected returns, which is analogous to the return predictability result of the price-to-earnings ratio in the equity market (e.g., Basu (1977); Campbell and Shiller (1988)).[8] High *pa* ratios are followed by low future returns in the cross-section of cryptocurrency returns. A zero-investment long-short strategy that buys the portfolio of coins with the lowest *pa* ratio and shorts the portfolio of coins with the highest *pa* ratio generates an average weekly excess return of 1.9%. We show that the return predictability of the *pa* ratio is distinct from those documented in the literature. Controlling for the cryptocurrency 3-factor model (see, Liu et al. (2022)), the alpha of the long-

[8]Several recent papers document empirical regularities related to cryptocurrency investment. Liu and Tsyvinski (2021) and Liu et al. (2022) comprehensively study the valuations of the cryptocurrency market in the aggregate time-series and the cross-section, respectively. Makarov and Schoar (2020), Borri and Shakhnov (2018), and Borri and Shakhnov (2019) find that there is dispersion of Bitcoin prices across different exchange platforms. Bhambhwani et al. (2020) show that cryptocurrency exposures to hash rate and network size growth predict cryptocurrency future returns. Contemporary works of Cong et al. (2021c) and Liebi (2022) provide other angles on studying the value premium of cryptocurrencies by considering a broad spectrum of variables. Cong et al. (2021b) focus on the cryptocurrency staking economy and examine a carry trade strategy.

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short strategy remains positive and statistically significant at the 1-percent level. We refer to this return predictability of the *pa* ratio as the cryptocurrency value effect, and explore potential mechanisms behind this value effect. We do not find evidence in support of several risk-based explanations and find some evidence potentially in line with the overreaction-based explanation of value effect.

Our paper adds to the growing literature on the role of disclosure in the cryptocurrency market. Prior studies have primarily focused on the effect of voluntary disclosure variables (e.g., white papers) during the ICO process. For example, Bourveau et al. (2022) comprehensively examine various voluntary disclosure items in the cryptocurrency market and their effects on the ICO market. Other studies also examine some aspects of voluntary disclosure during the ICO process, including the presence of white papers (e.g., Howell et al. (2020)), the content of white papers (e.g., Feng et al. (2019); Liu et al. (2019)), social media disclosures (e.g., Benedetti and Kostovetsky (2021)), and code disclosures (e.g., Adhami et al. (2018)). Broadly speaking, this paper relates to research on disclosures in alternative markets, such as the OTC market (e.g., Bushee and Leuz (2005)), the equity crowdfunding market (e.g., Cascino et al. (2019)), the peer-to-peer lending market (e.g., Michels (2012)), and the SPAC market (e.g., Chapman et al. (2021)). Relative to the literature, our study is different in at least two important ways. First, we examine the information content of disclosures in the secondary cryptocurrency market, rather than the primary market (i.e., ICO).[9] Second, the "accounting information" we examine is neither voluntary, as it does not have a discretionary component (e.g., white papers), nor regulation-mandated, as the cryptocurrency market is largely unregulated. Rather, this disclosure is unique in its transparent and verifiable nature, which is enabled by blockchain technology. Investigating the information content of such disclosure would foster a fruitful discussion of the information environment in the cryptocurrency market and our understanding of different disclosure types (e.g., Leuz and Wysocki (2016)).

Second, this paper relates to the literature studying the information content of accounting data in non-traditional accounting settings. For example, Amiram et al. (2022) show that data on

[9] As discussed in Bourveau et al. (2022), the lack of an ongoing disclosure regime during the ICO stage makes it difficult to assess the long-term performance of the coins.

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blockchain transaction volume can be used to predict the flow of terrorist funds on the blockchain system. Hand (2005) finds that financial statements are highly value-relevant in the venture capital market for a sample of U.S. biotechnology firms. Importantly, a large literature uses the value-relevance framework to study the information content of accounting data. Classic studies can be traced back at least to Beaver (1968), Ball and Brown (1968), and Fama et al. (1969). In particular, studies examining the value-relevance of earnings surprises find that equity returns can both lead earnings disclosure, known as pre-announcement drifts (e.g., Ball and Brown (1968)), and lag the disclosure, known as post-announcement drifts (e.g., Ball and Brown (1968); Bernard and Thomas (1989)). Using accounting methods to study the value-relevance of accounting data on the blockchain to investors, we find that information disclosed on new addresses is highly value-relevant in the cryptocurrency market. Moreover, we show that, contrary to the equity market, returns of cryptocurrencies do not lead or lag the disclosure of new address information, highlighting the distinct features of the information environment in the cryptocurrency market.

Third, this paper adds to the literature on understanding cryptocurrency valuation. The importance of user adoption is a foundational concept in the literature on network economy (e.g., Rohlfs (1974); Katz and Shapiro (1985); Katz and Shapiro (1994); Rochet and Tirole (2003)). Recent theoretical development of cryptocurrency valuation also emphasizes the role of user adoption and network effect. Cong et al. (2021a) present a model of dynamic user adoption, and their main finding is on the endogenous adoption of cryptocurrency and price predictions at different stages of the life cycles of cryptocurrencies. Sockin and Xiong (2020) link cryptocurrency prices to network sizes and show that network externalities in the market can lead to multiple equilibria. Other theoretical studies that focus on user adoption of cryptocurrencies include Pagnotta and Buraschi (2018), Biais et al. (2020), and Cong et al. (2022). Our paper provides comprehensive empirical analyses that highlight the importance of network effects in determining cryptocurrency value.

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II Network Economy and Cryptocurrency Value

In this section, we build on a large body of literature on network economy (e.g., Rohlfs (1974); Katz and Shapiro (1985); Katz and Shapiro (1986); Farrell and Saloner (1986); Katz and Shapiro (1994)) to discuss its relevance for valuing cryptocurrencies. The basic idea is that the value of a cryptocurrency increases when more users join its blockchain, which is commonly referred to as the network effect.

In a network economy, the utility a user derives from service depends on the overall user adoption of the economy, and therefore the valuation of the network increases when more users join. Accordingly, the cryptocurrency market can be considered a network economy.[10] The theoretical literature on cryptocurrency valuations also focuses on user adoption and network effect. Cong et al. (2021a) relate cryptocurrency prices to the user base. In a model of dynamic user adoption, they show that cryptocurrency prices are ultimately linked to the endogenous adoption of cryptocurrencies. Sockin and Xiong (2020) feature the network effect in their model and consider the possibility of multiple equilibria. Other studies that relate the network effect to cryptocurrencies include Pagnotta and Buraschi (2018), Biais et al. (2020), Pagnotta (2022), and Cong et al. (2022).

Empirically, studies show network effects in traditional network economies, such as software, ATM, and yellow pages (e.g., Gandal (1994); Saloner and Shepard (1995); Rysman (2004)). In particular, the value of internet companies, which are considered a classic example of network economy, is found to be strongly increasing when more users join. Trueman et al. (2000), Rajgopal et al. (2003), and Luo et al. (2013) find that the value of internet companies has a strong and positive relationship with their user bases. Importantly, the literature shows that traditional accounting numbers, such as earnings, are not related to the value of these internet firms (e.g., Govindarajan et al. (2018)). In other words, information of the user base is highly value-relevant for internet firms, while traditional accounting information is not.

[10] In principle, all cryptocurrencies have the functionality of payment systems even though a significant proportion of them is used primarily for non-payment purposes. Many of these cryptocurrencies also have platforms that serve specific purposes, such as storage, financing, and gaming.

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Both the network economy and the cryptocurrency research suggest a critical role of the network effect induced by user adoption. Based on the literature, the utility a cryptocurrency user derives from using the coin depends on the overall level of user adoption of the system, and therefore the valuation of the cryptocurrency would increase when more users join. Accordingly, we seek to empirically examine the usefulness and relevance of information regarding user adoption in the cryptocurrency market. To this end, we use the number of new addresses on blockchain to measure user adoption of cryptocurrencies, and employ the set of methods that are foundational in the valuation literature—the value-relevance tests. In capital market research, disclosed information (e.g., financial statements) is said to be value-relevant if they are associated with equity prices, values, or returns. The objective of the value-relevance tests is to examine whether and how quickly certain disclosed information, such as firm earnings, captures changes in the information set reflected in security returns over a specified period. These tests form the foundation of the extensive literature that empirically establishes the value-relevance of earnings in the equity market.[11]

The approach in this literature is to test how investors react to the disclosure of information, such as earnings. A common method is to regress contemporaneous stock returns on earnings surprises. The slope coefficients from these regression specifications are one of the important objects in the value-relevance literature (see, e.g., Dechow et al. (2010)). Moreover, the explanatory power, or adjusted R-squared of the regressions, measures the degree of informativeness of the information.[12]

Besides earnings, the accounting and finance literature also examines the value relevance of other financial statement information as well as non-accounting information using similar approaches (see surveys of Richardson et al. (2010) and Lewellen (2010) for discussions). For example, Amir and Lev (1996) study the value-relevance of potential user adoption in the wireless communication industry. Piotroski (2000) examines the role of historical financial statement information in predicting

[11] For related surveys of this literature, see Lev (1989), Fama (1998), Kothari (2001), and Scott and O'Brien (2019).

[12] Another method to test the usefulness of earnings is to examine the reactions of trading volume or return volatility to earnings news (e.g., Beaver (1968); Bamber (1987)). We also use this approach in Table A.1 in the Online Appendix, and reach similar conclusions. In this paper, we focus on market price reaction because it is easy to interpret and is believed to provide a stronger test for the decision usefulness than volume or return volatility reaction (see Kim and Verrecchia (1991)).

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stock returns. Hand (2005) examines the value relevance of financial statement data and non-financial statement information in the pre-IPO venture capital market. Patatoukas et al. (2015) study the value relevance of supplementary information in financial statements.

III Data Summary

We obtain on-chain and price data of cryptocurrencies from Glassnode.com, a major blockchain data and intelligence provider.[13] For each coin, Glassnode gathers its data recorded in the public ledger or blockchain, which is the record-keeping system of the user information, including identities, balances, and network transactions. The blockchain data forms the basis of our analyses. Because each cryptocurrency has both common and unique blockchain features, the number of available on-chain measures varies across the coins. Our main measures are based on the common features of blockchains and thus are available for the maximum amount of coins. Most of the on-chain data is available on a daily basis, and thus for our main empirical analyses, we construct the on-chain measures at the weekly frequency, which we discuss next.

To measure the information disclosed on new users of a blockchain, we use the number of new addresses per coin (NA). An address is a unique sequence of numbers and letters that represents a source or a destination of a blockchain payment. A new address is a unique address that appears for the first time in a transaction in a week. A high NA for a coin indicates a high user adoption of this coin. We then construct the weekly changes in logged NA, or Δna, to measure the incremental information, or surprises, embedded in the disclosed number of new addresses over a week.[14] We also construct the price-to-new address ratio, or pa ratio, defined as $pa = log(\frac{Price}{NA})$, where $Price$ is the end of day price on the last day of a week. The definitions of other variables are documented in the Appendix.

Our sample starts from February 2018, when the on-chain data became available for most coins,

[13] Other popular cryptocurrency on-chain data providers include Coinmetrics and IntoTheBlock. We use Glassnode because it provides the most comprehensive on-chain data among the popular data providers.

[14] We also construct and test alternative measures of the surprises in the disclosed number of new addresses in Section IV.1.

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to June 2021. We exclude stablecoins and require the coins to have price data. The final sample consists of 127 cryptocurrencies. Panel A of Table 1 shows the summary statistics of the main variables used in the paper. During the sample period, the average logged coin return is 0.3% per week and the weekly standard deviation of the logged coin returns is 0.251. The mean (median) growth in new addresses (Δna) is -1.9% (-3.2%), suggesting that the increase in the number of new users slightly slows down on average during our sample period. The summary statistics of the other variables are also included in Panel A. Panel B shows the correlation matrix of the variables. It shows that cryptocurrency returns are most correlated with the changes in the logged number of new addresses (Δna), with a correlation of 0.252. The correlations between cryptocurrency returns and the other variables are much lower. Table A.2 in the Online Appendix shows the summary statistics of the other variables used in the paper.

IV Value Relevance

Building on the discussion in Section II, we examine the value relevance of information disclosed on new addresses in the cryptocurrency market. In other words, we test if cryptocurrency returns react to the information disclosed on new addresses. One important concern of the value-relevance studies is to account for market expectations of the information disclosed (see, e.g., Lev (1989) and Kothari (2001)). In principle, only new information to the market would induce price reactions, and thus we need to measure the *surprises* in the new address information. In the returns/earnings relation studies in capital market research, surprises in disclosed earnings are measured as the difference between current earnings and a measure of its expected value. Three methods are commonly employed to estimate earnings expectations (see, e.g., Scott and O'Brien (2019)). The first method uses earnings of the previous period as the earnings expectations for the current period. This approach implicitly imposes an assumption that earnings are a random walk, and thus earnings growth is the appropriate measure of earnings surprises. The second method uses statistical models to calculate earnings expectations. The third approach uses analysts' forecasts of earnings as the

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market expectations. In the cryptocurrency market, both the first and second methods can be employed to construct proxies for market expectations of new addresses. Because of the lack of analysts' forecasts of new addresses in the cryptocurrency market, the third method is not readily applicable.

In our baseline specification, we use the changes in the logged number of new addresses to capture the surprises in the disclosed number of new addresses. The implicit assumption is that the cryptocurrency market uses the number of disclosed new addresses of the previous period as the expected new addresses for the current period, corresponding to the first method. In later analyses, we also use the second method to capture market expectations.

IV.1 Baseline Results

In the baseline regressions, we test whether cryptocurrency prices react to the surprises in the new address information in the following form:

$$r_{i,t} = \alpha + \beta \times \Delta na_{i,t} + \epsilon_{i,t} \tag{1}$$

where $r_{i,t}$ is the logged return of cryptocurrency i at time t, $\Delta na_{i,t}$ is the change in the logged number of new addresses of cryptocurrency i at time t, and $\epsilon_{i,t}$ is the residual term. The coefficient estimate of interest is β, which measures the price reactions of cryptocurrency i to the changes in its logged number of new addresses. The explanatory power, or adjusted R^2 of the regression, measures the degree of informativeness of the information. We use the Fama and MacBeth (1973) regression to study the relationship, which focuses on the cross-sectional variations.

Panel A of Table 2 presents the results. We run model (1) for each week, and report the average regression coefficient, the associated Fama and MacBeth (1973) t-statistic, and the average adjusted R-squared. In column (1), the coefficient estimate of Δna, as the only independent variable, is positive and highly statistically significant, suggesting that the cryptocurrency market strongly and positively responds to the information contained in Δna. A one-standard-deviation

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increase in Δna is associated with a 4.9% increase in logged returns. Δna alone already explains 8.5% of the variations in cryptocurrency returns, indicating that information about new addresses is highly value-relevant in the cryptocurrency market. Column (2) includes na as an additional explanatory variable to capture the level effects. The coefficient estimate of na is positive and statistically significant at the 10-percent level. The adjusted R^2 increases slightly to 8.8% from 8.5% of the standalone regression.[15]

IV.1.1 Alternative Measures of Market Expectation

The baseline approach assumes that the cryptocurrency market uses the number of disclosed new addresses of the previous period as the new address expectation for the current period, and thus imposes a relatively strong assumption that the number of new addresses is a random walk or at least the investors believe so. We relax this assumption and use alternative statistical models to estimate market expectations for the number of new addresses.

We use two statistical methods: (1) the cross-sectional and (2) the time-series methods. For the cross-sectional method, we first regress the logged number of new addresses of cryptocurrencies at time $t-1$ on their logged number of new addresses at time $t-2$ and keep the coefficient estimates. Then, we multiply the coefficient estimates with the logged number of new addresses at time $t-1$ to calculate the predicted logged number of new addresses at time t, which we use as the expected logged number of new addresses. Therefore, the surprises in the disclosed number of new addresses are the differences between the realized logged number of new addresses and the predicted logged number of new addresses, which we denote as $na\ shock^1$.

To calculate the expectations of new addresses using the time-series method, we first regress the logged number of new addresses of cryptocurrency i on its logged number of new addresses in the previous period over the 12 weeks proceeding the current period, and keep the coefficient estimates.[16] Then, we apply the coefficient estimates to the logged number of new addresses for

[15] In Table A.3 of the Online Appendix, we show that results stay similar when we only use ERC20 cryptocurrencies, or the cryptocurrencies built on the Ethereum blockchain.

[16] We use a 12-week window to conduct all rolling estimations in the paper. While there is no consensus on the estimation window, requiring 12 weeks for rolling estimations in the cryptocurrency market balances precision and

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cryptocurrency i at time $t-1$ to calculate the predicted logged number of new addresses at time t, which we use as the expected logged number of new addresses. Similarly, the surprises in the disclosed number of new addresses are the differences between the realized logged number of new addresses and the predicted logged number of new addresses, which we denote as $na\ shock^2$.

Panel B and Panel C of Table 2 document the results based on $na\ shock^1$ and $na\ shock^2$. We focus our discussion on $na\ shock^1$. The results for $na\ shock^2$ are qualitatively similar. In the standalone model in column (1) of Panel B, the coefficient estimate of $na\ shock^1$ is positive and highly statistically significant. The point estimate is 0.085, which is close in magnitude to that of Δna at 0.084 in Panel A. The adjusted R^2 is 8.6%, which is also similar to that based on Δna at 8.5%. Column (2) of Panel B further includes na as an explanatory variable. Interestingly, the point estimate of na is no longer significant when $na\ shock^1$ is used instead of Δna. This result suggests that the cryptocurrency market does not respond to na when $na\ shock^1$ is available.

IV.1.2 Controlling for Volume, Attention, and Supply

An important concern of our exercise is that the growth of the number of disclosed new addresses may capture changes of investor demand or attention for a cryptocurrency. To mitigate the concern, we control for proxies of investor demand using the actual transaction volume of cryptocurrencies and investor attention using Google searches. Table 3 documents the results. The coefficient estimates of Δna are stable and remain highly statistically significant controlling for changes in logged transaction volume and changes in logged Google searches, suggesting that the value-relevance of the number of new addresses is unlikely to be driven by these factors. The adjusted $R^2 s$ increase after including the additional variables, but the increases relative to the baseline specification are small. Another concern is that the value-relevance of new addresses is driven by the supply of coins. We explicitly control for the changes in the logged total amount of circulating supply of the coins in the model ($\Delta current$). The point estimates on Δna remain stable when we include the additional variables. Furthermore, we control for changes in the

potential time-varying coefficient estimates. Using longer windows (24 weeks or 36 weeks) generates qualitatively similar results.

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logged percentage of supply held by the top 1% addresses ($\Delta balance1$) and changes in the logged total amount of coins held by exchange addresses ($\Delta balance2$), to account for the effects of large holders and exchanges. Overall, the results show a strong returns/new-addresses relation in the cross-section of cryptocurrencies and that the value-relevance of Δna is not subsumed by investor demand, attention, the circulating supply of the coins, or influential holders.

IV.1.3 Adjusting for Expected Returns

In the previous analyses, we use raw returns to measure cryptocurrency market reactions. However, the raw returns may capture returns reaction to both coin-specific news and cryptocurrency market-wide news, the latter of which we want to remove. In capital market research, the common method to remove the component of expected returns is to calculate the residual returns using benchmark factor models.[17] The idea is that factor models capture the expected component of returns, and therefore the residual returns would only contain information that is orthogonal to the expected returns, or the coin-specific news.

We apply this method to the cryptocurrency market using the cryptocurrency CAPM and cryptocurrency 3-factor models as in Liu et al. (2022). Liu et al. (2022) show that a cryptocurrency three-factor model including the cryptocurrency market factor, cryptocurrency size factor, and cryptocurrency momentum factor is able to capture the cross-section of expected cryptocurrency returns. Therefore, we use the cryptocurrency factor models to remove components that are attributable to the expected returns. We define residual returns as the difference between raw returns and factor model implied expected returns:

$$res_{it} = R_{it} - \left(\hat{\alpha_{it}} + \hat{\beta}'_{ikt} R_{kt}\right)$$

where R_{kt} is either the cryptocurrency market factor or the three factors including the cryptocurrency market factor, size factor, and momentum factor; $\hat{\alpha}_i$ and $\hat{\beta}_{ik}$ are estimated from the 12 weeks

[17] Removing the component of expected returns is a common practice in the literature to separate market-wide effect from firm-specific effect and is used by most existing value relevance studies. see Lev (1989) and Kothari (2001) for detailed discussions on the merits of using different factor models to capture components of expected returns.

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preceding the current period; i indicates the asset, t indicates time, and k indicates the factor.

The results are documented in Table 4. Panel A shows the results adjusting for the cryptocurrency CAPM model. The point estimates of Δna are positive and highly statistically significant. The adjusted R^2 for the specification using only Δna is 7.7%, which is similar to the specification using raw returns at 8.5%. Panel B reports the results adjusting for the cryptocurrency 3-factor model. The point estimates of Δna are positive and highly statistically significant. The adjusted R^2 for the specification using only Δna is 5.8%. The coefficient estimates of na are neither positive nor significant in any of the specifications. That is, the cryptocurrency market responds strongly to the surprises in the disclosed number of new addresses but not to the simple amount of new addresses. Overall, the results based on unexpected returns further support our baseline results that the disclosed number of new addresses is highly value-relevant in the cryptocurrency market.

IV.2 Economic Determinants of Value-Relevance

In the equity market, studies have shown several important determinants of the returns/earnings relation (see, e.g., Kothari (2001); Dechow et al. (2010)). The first type of determinant is the quality of earnings, measured using both the statistical property of earnings and the property of the accounting system. The second main determinant is the discount rate. The third one is firm size, commonly used as a measure of the informativeness of stock prices. The three types of determinants in the returns/earnings relation correspond to the three components of the present value discounting formula – cash-flow (earnings), discount rate, and stock price. Motivated by the evidence in the equity market and the theoretical predictions regarding cryptocurrency prices (e.g., Cong et al. (2022); Biais et al. (2020)), in this subsection, we construct the counterparts of the potential economic determinants for the returns/new-addresses relation for cryptocurrencies.

Persistence and Variability

Research in equities shows that the quality of earnings can differ across companies (e.g., Dechow et al. (2010)). If the quality is high, one may expect the value-relevance of earnings

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to also be high since investors are better able to predict future firm performance using current earnings. There are two main approaches to measuring earnings quality. The first is based on the statistical property of earnings, and the second is based on the attributes of the accounting system.

For the first approach, one of the main measures is earnings persistence. The idea is that the value-relevance of earnings should be high if the good or bad news in current earnings is expected to persist into the future. Kormendi and Lipe (1987), Basu (1997), and Li (2011) propose and confirm that high earnings persistence is linked to high returns/earnings relation in the equity market. Another measure of earnings quality based on its statistical property is earnings variability. It is proposed that investors prefer earnings with less variability as it suggests low volatility of the underlying business model (e.g., Schipper and Vincent (2003)). For the second approach, research finds that accounting systems can affect the value-relevance of earnings. For example, Leuz and Verrecchia (2000), Bailey et al. (2006) and Greenstone et al. (2006) show that stringent accounting standards increase the value-relevance of earnings. The statistical properties based on persistence and variability are readily applicable to our setting, whereas the accounting system is not. Accordingly, we focus on studying the role of persistence and variability of new address growth in the cryptocurrency market.

We first test the relation between persistence and the value-relevance of the disclosed number of new addresses. For each period, we split the cross-section of cryptocurrencies into two groups based on the persistence of Δna in the past 12 weeks, where the persistence is estimated as the point estimate of regressing Δna on its lagged value. We construct an indicator variable $1_{\{p>Med\}}$ which equals one if the persistence of Δna is above the sample median and zero otherwise. Columns (1) and (2) of Panel A in Table 5 show the results for the subsamples of coins with below and above median persistence of Δna, respectively. The point estimate of Δna is 0.105 for the above-median subsample, while the point estimate of Δna is 0.077 for the below-median subsample. That is, the sensitivity between returns and new address growth is about 30% higher for the subsample with high new address growth persistence. Column (3) reports the results based on the full sample and includes the cross term between $1_{\{p>Med\}}$ and Δna. The coefficient estimate is positive and

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statistically significant at the 5-percent level, confirming that the returns/new addresses relation is more pronounced among the coins with above-median persistence of Δna.

Next, we study the relationship between the variability of new address growth and the value-relevance of new addresses. For each period, we split the cross-section of cryptocurrencies into two groups based on the variability of new address growth, measured as the standard deviation of Δna in the past 12 weeks. We construct an indicator variable $1_{\{v>Med\}}$ which equals one if the variability of Δna is above the sample median and zero otherwise. Columns (1) and (2) of Panel B in Table 5 show the results for the subsamples of coins with below and above median variability of Δna, respectively. The point estimate of Δna is 0.073 for the above-median subsample, while the point estimate of Δna is 0.123 for the below-median subsample. Column (3) reports the results based on the full sample and includes the cross term between $1_{\{v>Med\}}$ and Δna. The coefficient estimate is negative and statistically significant at the 1-percent level, confirming that the returns/new addresses relation is more pronounced among the coins with below median variability of Δna.

In summary, we find that the returns/new addresses relation is more pronounced among coins with a high persistence of Δna and with a low variability of Δna.

Discount rate

Another potential determinant of the value-relevance of earnings is the discount rate. When the discount rate is high for a stock, the discounted present value of the revisions in expected future earnings would be low. That is, the value relevance of earnings is expected to be lower for companies with high discount rates. Empirically, prior studies test and confirm the negative relationship between discount rates, including the risk-free and risky components, and the value relevance of earnings (e.g., Collins and Kothari (1989); Easton and Zmijewski (1989)).

Similarly, if the cryptocurrency valuation also follows a present value discounted formula (e.g., Cong et al. (2022); Biais et al. (2020)), we would expect to find a relatively low value relevance of new addresses when the discount rate is high for a cryptocurrency. To measure the discount rate of cryptocurrencies, we use the cryptocurrency 3-factor model as in Liu et al. (2022). For each

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cryptocurrency, we estimate the factor loadings on the three factors with a rolling 12-week window preceding the current period. The implied discount rate is calculated based on the following formula:

$$\beta_{i,t}^{CMKT} E\left[R^{CMKT}\right] + \beta_{i,t}^{CSMB} E\left[R^{CSMB}\right] + \beta_{i,t}^{CMOM} E\left[R^{CMOM}\right] + R^{f}$$

where $E\left[R^{CMKT}\right]$, $E\left[R^{CSMB}\right]$, and $E\left[R^{CMOM}\right]$ are calculated as the average excess returns of the factors from Liu et al. (2022). Table A.4 in the Online Appendix shows the results. Overall, we find that, consistent with the argument above, the sensitivity between returns and new addresses growth is about 15% higher for the low discount rate subsample. However, the difference is not statistically significant.

Size

In the equity market, it is theoretically proposed that the informativeness of stock prices, often proxied by firm size, may be negatively related to the value relevance of earnings (e.g., Easton and Zmijewski (1989); Collins and Kothari (1989); Kothari (2001)). The argument is that large firms tend to receive more attention and are under more scrutiny by investors, and thus there is less information content in the current accounting earnings, leading to a lower value-relevance of earnings at the disclosure of information.[18] A similar argument may be applied to the cryptocurrency market, suggesting a negative relation between size and the value-relevance of new addresses.

On the other hand, there could be a positive relationship between size and the value relevance of new addresses for cryptocurrencies. In the theoretical literature of both the network economy and the cryptocurrencies, models with network externality (e.g., Rohlfs (1974); Katz and Shapiro (1985); Armstrong (2006); Cong et al. (2021a)) imply a potentially positive relationship. In studying the internet companies in the equity market, Rajgopal et al. (2003) show that the value relevance of the number of website users is higher for large e-commerce companies, and they argue

[18]The empirical evidence on the relationship between size and the value-relevance of earnings is mixed (e.g., Easton and Zmijewski (1989); Collins and Kothari (1989)).

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that this non-linearity is due to network effects.

To test whether size is a determinant of the value-relevance of new addresses, for each period, we split the cross-section of cryptocurrencies into two groups based on their network sizes measured as the total number of addresses.[19] We construct an indicator variable $1_{\{A>Med\}}$ which equals one if the total number of address is above the sample median in the previous period and zero otherwise. Table 6 documents the results. Columns (1) and (2) show the results for the subsamples of coins with below-median network size and above-median network size, respectively. The point estimate of Δna is 0.080 for the below-median subsample, while the point estimate of Δna is 0.103 for the above-median subsample. That is, the sensitivity between returns and new addresses growth is about 29% higher for the large network size subsample. Column (3) reports the results based on the full sample and includes the cross term between $1_{\{A>Med\}}$ and Δna. The coefficient estimate of the cross term is positive and statistically significant at the 5-percent level. Overall, we find that the returns/new addresses relation is more pronounced among coins with larger network sizes, consistent with the theoretical argument regarding network effects. The results also highlight some differences between the equity and cryptocurrency markets.

V Discrete and Continuous Revelation of Information

An important difference between cryptocurrencies and equities is that, for cryptocurrencies, information of the underlying economic activities on blockchain is disclosed publicly and (almost) continuously. This is in striking contrast to the disclosure of earnings and other financial information of publicly traded companies. Under the current reporting standards, public companies are required to report three quarterly earnings and one annual earnings.[20]

This discrete nature of corporate earnings is extensively studied in the accounting and finance

[19] Using market cap generates qualitatively similar results

[20] This quarterly reporting system was not established until 1970. Before the Securities Act of 1933 and the Securities Act of 1934 (Securities Acts), stock exchanges largely governed financial reporting, and there was no homogeneous reporting requirement across exchanges. Since the passage of the Securities Acts, the SEC has played a primary role in regulating the financial reporting of public companies. Specifically, the SEC mandated annual reporting in 1934 and semiannual reporting in 1955. In 1970, the SEC established the current quarterly reporting system.

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literature. Theoretically, the net effect of reporting frequency is unclear (e.g., Diamond (1985); Diamond and Verrecchia (1991); Abreu et al. (1991)). Empirical studies examine both the benefits and costs associated with more frequent reporting. On the benefit side, studies focus on the additional transparency that more frequent reporting brings. Butler et al. (2007) show that stock prices reflect accounting information more quickly when firms voluntarily increase their reporting frequencies. More frequent reporting is also found to be associated with reduced cost of capital and improved liquidity (e.g., Fu et al. (2012)). On the cost side, both theories (e.g., Gigler et al. (2014)) and empirical studies suggest that higher reporting frequencies can cause managers to myopically prefer short-term performance gains at the expense of overall firm value. Kraft et al. (2018) claim that increased reporting frequency is associated with a large decline in investments, consistent with managers behaving myopically following increases in reporting frequency. Ernstberger et al. (2017) show that firms with more frequent reporting exhibit higher real earnings management.

Two important phenomena in the equity market are attributable to the infrequent disclosure of financial reporting information and suggest that information is not fully efficiently incorporated into prices. The first phenomenon is referred to as the post-announcement drift. That is, information, such as unexpected earnings, tends to forecast future company returns (e.g., Ball and Brown (1968); Bernard and Thomas (1989)). Among many alternative explanations for this phenomenon, a common view is that the equity market is slow in incorporating earnings information. The second phenomenon is referred to as the pre-announcement drift. That is, stock returns tend to precede informational events such as earnings announcements (e.g., Ball and Brown (1968)). The literature commonly attributes the phenomena to the discreteness of information disclosure and information leakages between the discrete announcements, respectively.

For cryptocurrencies, the accounting information disclosed on blockchain is made public and nearly continuous by design. Thus, the cryptocurrency market provides an ideal setting to study frequent reporting where value-relevant information is revealed almost continuously. In this section, we revisit these classic information disclosure-related phenomena in the cryptocurrency market.

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V.1 Slow Acquisition of Information

Post-earnings announcement drift (PEAD), first documented by Ball and Brown (1968), is one of the earliest and most well-known anomalies in the literature. The PEAD phenomenon can be described as that after earnings are announced, estimated cumulative excess returns drift up for positive earnings surprises (good news) and drift down for negative earnings surprises (bad news). A common explanation of PEAD is the slow acquisition of information (e.g., Bernard and Thomas (1989)).

In the cryptocurrency market, information disclosed on new addresses is publicly and continuously available. This environment provides a setting to study investors' acquisition of information where the value-relevant information can be obtained at any time. However, it is ex ante unclear whether such an environment would facilitate investors' information acquisition. On the one hand, attention can be a scarce resource (e.g., Peng and Xiong (2006)) and continuous revelation of information can lead to severe under-reaction from investors.[21] On the other hand, it is shown that stock prices reflect accounting information more quickly when firms voluntarily increase their reporting frequencies (e.g., Butler et al. (2007)).

To study the information acquisition of news pertaining to new addresses in the cryptocurrency market, we use Δna to predict subsequent cryptocurrency returns. The results are documented in Table 7. Columns (1) to (3) show the results of using Δna to predict one-, two-, and three-week ahead logged returns, respectively. The point estimates of Δna for the one-, two-, and three-week ahead predictability results are -0.005, 0.005, and -0.005, respectively, which are insignificant and small compared to the contemporaneous response coefficients. In columns (4) to (6), we further include the cryptocurrency beta, size and momentum characteristics as controls.[22] The point estimates of Δna remain insignificant and the magnitudes of the coefficients become even smaller. The evidence suggests that the cryptocurrency market is efficient in responding to the surprises in the disclosed number of new addresses, consistent with the idea that frequent revelation

[21] Another plausible explanation of PEAD is based on changes in uncertainty and risks (e.g., Sadka (2006); Francis et al. (2007)).

[22] Beta is measured as the exposure of individual return to cryptocurrency market return in the past 12 weeks.

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of information facilitates information acquisition.

V.2 Information Leakage

Next, we examine the potential presence of return drifts prior to the disclosure of new address information. In the equity market, prior studies find that stock returns can precede earnings surprises at earnings announcements by weeks or even months (e.g., Ball and Brown (1968)). That is, firms with positive earnings surprises tend to outperform firms with negative earnings surprises even prior to the disclosure date of earnings. A common explanation for the pre-drift phenomenon is that, because of the discreteness of the events, information gradually leaks to the market prior to the disclosure date (e.g., Collins and Kothari (1989)). For cryptocurrencies, the disclosure of blockchain information is almost continuously, and thus any potential information leakage, which depends on the discreteness of disclosure, may not be important in the cryptocurrency market.

To test whether the pre-drift phenomenon is present in the cryptocurrency market, we regress future changes in the logged number of new addresses on the current logged returns. The results are summarized in Table 8, which uses logged returns to predict future Δna up to three weeks ahead. The results show that cryptocurrency returns do not predict future Δna at the one-week and two-week horizons. At the three-week horizon, cryptocurrency returns predict Δna significantly, but the coefficient estimate is negative, which is inconsistent with the information leakage hypothesis. The economic magnitude is small–a one-standard-deviation (25.1%) increase in ret is only associated with a 1.8% increase in Δna. Overall, we conclude that returns do not immediately precede the disclosure of new accounting information in the cryptocurrency market, plausibly due to the fact that the information is disclosed publicly and almost continuously in this market.

V.3 Cumulative Return around New Address Information

To further understand the behavior of cryptocurrency returns around the disclosure of new address information, we summarize the evidence in Figure 1. Specifically, we plot the cumulative average logged abnormal returns from six weeks before to six weeks after the disclosure of new

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address information for subgroups of cryptocurrencies based on positive and negative new address growth.

Consistent with the evidence so far, there is a significant market reaction to new address information on the event date. However, both pre- and post- drifts are muted. The return behavior in the cryptocurrency market is in sharp contrast to the return behavior of the equity market around earnings disclosure. Figure A.1 in the Online Appendix shows the celebrated results of the pre- and post-announcement drifts around earnings announcements from Ball and Brown (1968). The figure shows a strong pre-announcement drift even 12 months before the announcement date and a post-announcement drift that persists a few months after the announcement date. The presence of strong market reactions at the disclosure of new address information, and the absence of pre- and post-drifts around the disclosure highlight the distinct features of the information environment in the cryptocurrency market and provide a benchmark case where information is disclosed both publicly and (almost) continuously.

VI Price-to-New Address Ratio and Return Predictability

The price-to-earnings ratio is one of the most studied financial ratios for the equity market. To further explore the role of accounting information in the cryptocurrency market, in this section, we use the number of new addresses to construct a potential counterpart in the cryptocurrency market: the price-to-new address ratio (pa ratio). Similar to earnings per share, our new address measure of a cryptocurrency is scaled by its outstanding coins as discussed in Section III. We first describe its properties. Then, we show that it is a strong cross-sectional cryptocurrency return predictor and that its predictive power is distinct from the existing predictors such as the cryptocurrency market, size, and momentum factors. Lastly, we investigate the potential mechanism underlying the result.

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VI.1 The Price-to-New Address Ratio

We first show in Figure 2 a graphical illustration of the logged price-to-new address (pa) ratios of Bitcoin, Ethereum, and the sample average from 2018 to 2021. The pa ratios of Bitcoin and Ethereum are close to each other during the period. The average pa ratio for the full sample of coins (12.3) is higher than those of Bitcoin and Ethereum during most of the period. In the time series, the pa ratios are relatively low at the beginning of 2019 and 2020. Over the two years of 2020 and 2021, the pa ratios have been steadily increasing.

VI.2 Return Predictability

In the asset pricing literature, a large number of studies (e.g., Basu (1977); Campbell and Shiller (1988)) document that the price-to-earnings ratios negatively predict returns in the future. That is, when the price of a stock is high relative to its earnings, the stock tends to have lower returns in the future. Motivated by the evidence from the equity market, we examine the relationship between cryptocurrency expected returns and the price-to-new address ratios.

First, we analyze the performance of the zero-investment long-short strategy based on the pa ratios. Each week, we sort individual cryptocurrencies into quartile portfolios based on the value of pa ratios of the corresponding coins.[23] We track the return of each portfolio in the week that follows. Then, we calculate the average excess returns over the risk-free rate for each portfolio, as well as the excess returns of the long-short strategies based on the difference between the fourth and the first quartiles.

Table 9 presents the results based on portfolio sortings. Panel A of Table 9 shows the average excess returns, standard deviations, and Sharpe ratios of the quartile portfolios and the zero-investment long-short strategy. The average mean excess returns decrease monotonically from the lowest pa ratio portfolio to the highest pa ratio portfolio. The average mean excess return of the lowest pa ratio portfolio is 1.5% per week, and that of the highest pa ratio portfolio is -0.4%

[23] We choose quartile portfolios because the average number of cryptocurrencies in the cross-section is about one hundred. Therefore, quartile portfolios ensure that there are about 25 coins in each portfolio. In robustness tests, we find that sorting cryptocurrencies into tercile or quintile portfolios gives qualitatively similar results.

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per week. The difference in the average returns of the lowest quartile and the highest quartile is 1.9% per week, which is statistically significant at the 1-percent level. The Sharpe ratios also decrease monotonically from the lowest to the highest *pa* ratio portfolios. The weekly Sharpe ratio for the lowest *pa* ratio portfolio is 0.132 (0.951 annually), and that for the highest *pa* ratio portfolio is -0.038 (-0.274 annually). The weekly Sharpe ratio for the zero-investment long-short strategy is 0.279 (2.012 annually). We refer to the return predictability of the price-to-new address ratio as the cryptocurrency value effect.

Panel B of Table 9 reports results controlling for either the cryptocurrency CAPM model or the cryptocurrency 3-factor model (see Liu et al. (2022)). In the first part of Panel B, we use the cryptocurrency CAPM model, where cryptocurrency market excess returns are included. The beta loadings to the cryptocurrency market factor decrease monotonically from the lowest to the highest *pa* ratio portfolios. The zero-investment long-short strategy significantly loads on the cryptocurrency market factor. However, the alpha of the long-short strategy remains positive and significant at the 1-percent level. In the second part of Panel B, we use the cryptocurrency 3-factor model from Liu et al. (2022). The long-short strategy loads positively and significantly on the cryptocurrency market factor, while negatively and significantly on the cryptocurrency size factor. The long-short strategy does not significantly expose to the cryptocurrency momentum factor. The alpha of the long-short strategy remains positive and significant at the 1-percent level after controlling for the cryptocurrency 3-factor model. The economic magnitude of the alpha is 1.9% per week. These results show that the return predictability cannot be subsumed and is distinct from those of the cryptocurrency market, size, and momentum effects. In the Online Appendix, we present additional robustness checks for the return predictability results, including different horizons, the Fama-MacBeth method, and alternative portfolio formation.

VI.3 Investigating Mechanism

Next, we explore the potential mechanism for the return predictability of the price-to-new address ratio. The asset pricing literature has proposed various potential explanations for the return

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predictability related to price-to-fundamental ratios, such as price-to-earnings, price-to-dividend, and market value-to-fundamental value ratios. The explanations can be largely categorized into either risk-based or behavioral explanations.

Risk-based explanation

The literature has proposed at least three risk-based explanations. The first one is based on the distressed risk mechanism (e.g., Fama and French (1996)). The idea is that certain assets have an elevated probability of failing or persistently weak performance, referred to as distressed assets. The returns of these distressed assets tend to move together, so their risk cannot be diversified away, and thus investors demand a premium to bear the risk, leading to high average returns. However, Lakonishok et al. (1994) find that the strategies based on valuation ratios, which may capture relative distress, do not underperform during business downturns. Moreover, Campbell et al. (2008) directly measure the distress risks of companies and show that distressed companies have relatively low returns on average.

Motivated by this explanation, we test if cryptocurrencies with low price-to-new address ratios perform poorly during business or market downturns. In Table A.5 of the Online Appendix, we show the average daily cryptocurrency long-short strategy returns based on the pa ratio for each month, as well as the NBER recession indicator, an indicator that equals one if the cumulative excess return of the coin market is less than –20% in the month, and an indicator that equals one if the cumulative excess return of the stock market is less than –5% in the month. For the two NBER recession months, the average daily return of the strategy is slightly negative for March 2020, but strongly positive for April 2020. For the three months with large negative cumulative coin market excess returns and the five months with large negative cumulative stock market excess returns, there are no discernible patterns in the strategy excess returns. Our results in the cryptocurrency market are largely consistent with Lakonishok et al. (1994) for the equity market that the strategies based on valuation ratios do not underperform during business or market downturns.

Two other risk-based theories are commonly referred to in explaining the return predictability

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of the valuation ratios. The first theoretical mechanism is formalized by Zhang (2005). Zhang (2005) argues that costly reversibility makes it difficult to reduce assets in place, and thus growth (value) firms are less (more) risky in bad times when reducing assets is preferred. The second theoretical mechanism is based on the concept of investment-specific shocks, or *ist* shocks (e.g., Papanikolaou (2011); Kogan and Papanikolaou (2014)). It is argued that there are *ist* shocks that improve investment opportunities, which increase the representative household's marginal utility. The *ist* shocks benefit firms with more growth options, and increase the value of growth opportunities relative to that of existing assets, leading to the return predictability of valuation ratios. The important implications of these two theories are that the long-short strategy returns based on valuation ratios should comove with macroeconomic shocks.

In Table 10, we test the exposures of the cryptocurrency long-short strategy excess returns based on the *pa* ratio to several macroeconomic shocks. We use four different macroeconomic variables to capture shocks to the aggregate economy. The first three variables, including changes in consumption growth, durable consumption growth, and = industrial production growth, are designed to capture the first-order aggregate shocks. The last variable, changes in the logged investment-to-consumption ratio similar to Papanikolaou (2011) and Kogan and Papanikolaou (2014), is designed to capture the *ist* shocks. Table 10 shows that the excess returns of the long-short strategy based on the price-to-new address ratios do not significantly expose to these macroeconomic variables, inconsistent with the predictions of the two risk-based theories.

In summary, we show that the long-short strategy excess returns based on the price-to-new address ratio are not concentrated in business or market downturns. The strategy excess returns are also not significantly exposed to he macroeconomic shocks, including consumption growth, durable consumption growth, industrial production growth, and investment-specific shocks.

Behavioral explanation

The behavioral explanations of the return predictability of the valuation ratios are largely based on the idea of overreaction. The theories of overreaction state that investors may overreact to the

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news or performance of certain assets, leading to the predictability of valuation ratios (e.g., Daniel et al. (1998); Barberis et al. (2015)).

An implication of the overreaction explanation is that assets with high valuation ratios and low growths tend to underperform, because these assets are overpriced due to investor overreaction and would converge to the fundamental value over time. In line with the prediction, we study the interaction between the price-to-new address ratio and the address growth of the coins. Panel A of Table 11 reports the subsample results based on the coins' address growths. For each week, coins are first sorted into the low and high groups based on their address growths. Then, within each group, the coins are further sorted into four portfolios based on their price-to-new address ratios. We find that the return predictability of the pa ratio largely concentrates in the low address growth subsample. Within the low address growth subsample, the long-short strategy that longs the coins with low pa ratios and shorts the coins with high pa ratios generates an average excess return of 2.68% per week that is statistically significant at the 1-percent level. For the coins with high valuation ratios and low address growth, the average excess returns are -0.94% per week in the future. The subsample result is consistent with the notion that investors overreact to certain coins with low growth whose prices gradually converge to the correct value, leading to lower average returns for these coins.

Another implication that is common to behavioral theories is that the return predictability should be stronger among the assets with relatively high costs of arbitrage (e.g., Shleifer and Vishny (1997); Pontiff (2006); Atilgan et al. (2020)). Following Liu et al. (2022), we construct the cost of arbitrage index for each coin and test the long-short strategy in the subsamples based on the coins' cost of arbitrage indexes. A high index value corresponds to a high cost of arbitrage. For each week, coins are first sorted into the low and high groups based on the cost of arbitrage index. Then, within each group, the coins are further sorted into four portfolios based on their price-to-new address ratios. In Panel B of Table 11, we find that the average excess return is larger for the high cost-of-arbitrage subsample, relative to the low cost-of-arbitrage subsample, which is consistent with the implications of behavioral explanations. Overall, the evidence is potentially in

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line with the behavioral explanations of overreaction. We note that although we provide evidence to support some plausible mechanisms behind the return predictability of valuation ratios in the cryptocurrency market, the channels are possible mechanisms and are not conclusive.

VII Conclusion

Given the absence of a standardized accounting framework for cryptocurrencies, coupled with concerns around price volatility, it is important to explore alternative methods for valuing cryptocurrencies that extend beyond market-based information. Meanwhile, cryptocurrencies are the first asset class that discloses a complete history of economic activities recorded on the blockchain, providing a rich set of potentially reliable information for investors. In this paper, we show that the blockchain information disclosed can be used to assess the intrinsic value of cryptocurrencies. In particular, using the accounting value-relevance framework, we show that the information disclosed on user adoption of cryptocurrencies, as measured by the number of new addresses, is highly value-relevant, suggesting that investors significantly react to blockchain information disclosure. Moreover, we find evidence suggesting that investors incorporate disclosed blockchain information efficiently – both pre- and post-drifts are absent around the disclosure of new address information. The presence of strong market reactions at the disclosure of new address information and the absence of pre- and post-drifts highlight the distinct features of the information environment in the cryptocurrency market and provide a benchmark case where information is disclosed both publicly and continuously. Lastly, to further explore the role of accounting information in this market, we construct the price-to-new address ratio for each coin. We show that the price-to-new address ratios negatively predict future cryptocurrency returns in the cross-section–a cryptocurrency value effect.

Our paper adds to the growing literature on disclosure in the cryptocurrency market, as well as to the broader examination of the information content of disclosure in non-traditional accounting settings. The disclosure we examine is neither voluntary nor regulation-mandated, but unique in its transparent and verifiable nature enabled by blockchain technology. Our findings highlight the

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value of credible and verifiable information that is publicly and (almost) continuously available, especially in a fast-growing market that lacks adequate investor protection laws. Understanding the information content and value of this type of disclosure would foster a constructive discussion on the information environment in the cryptocurrency market. Furthermore, our evidence is relevant to policymakers, indicating that accounting information can be valuable for digital assets.

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Figure 1: **Cumulative Abnormal Return around New Address Information**

This figure plots the cumulative average logged abnormal return around the disclosure of new address information for positive and negative new address growth subgroups. Dashed lines represent the 5-percent confidence intervals.

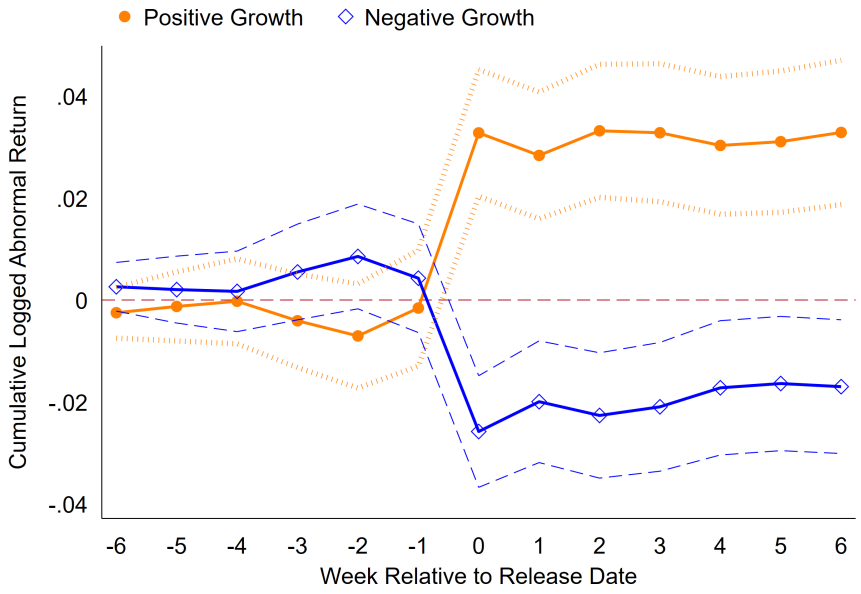
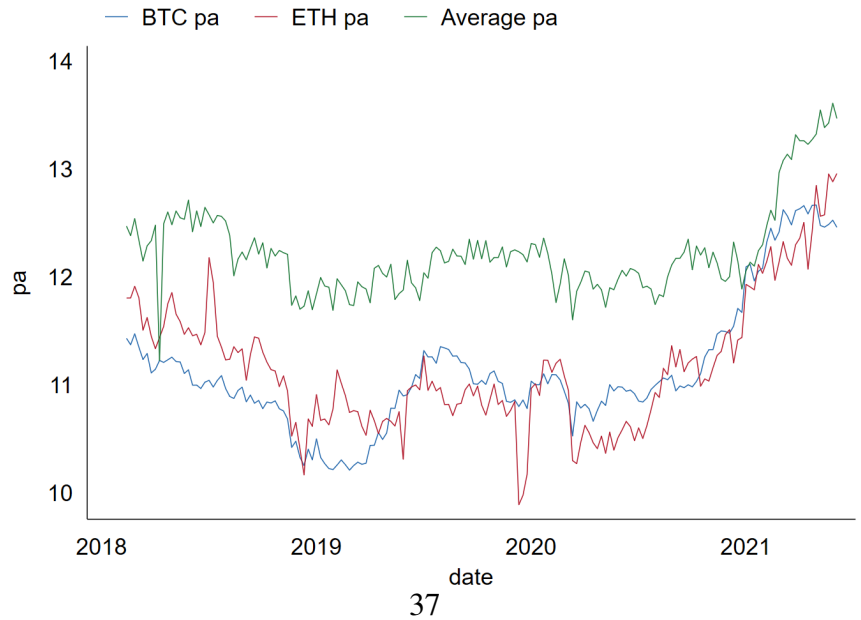


Figure 2: **Price-to-New Address Ratio of Major Coins**

This graph plots the logged price-to-new address ratio of Bitcoin, Ethereum, and and sample average from 2018 to 2021. The blue line, red line, and green line show the results for Bitcoin, Ethereum, and sample average of the logged price-to-new address ratio, respectively.



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Table 1: **Summary Statistics**

This table shows the summary statistics of the main variables used in the paper. Panel A reports the mean, standard deviation, 10% percentile, median and 90% percentile values of the variables. Panel B reports the correlation matrix of the variables.

Panel A	Mean	SD	10%	Median	90%
ret	0.003	0.251	-0.214	-0.000	0.210
Δna	-0.019	0.582	-0.551	-0.032	0.542
na	-13.095	12.362	-17.009	-13.449	-8.787
pa	12.254	1.344	10.844	12.114	13.902

Panel B	ret	Δna	na	pa
ret	1.000			
Δna	0.252	1.000		
na	0.039	0.083	1.000	
pa	0.004	-0.206	-0.167	1.000

Table 2: **Value Relevance Analysis**

This table shows the results for the cross-sectional value-relevance tests. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	Panel A		**Panel B**		**Panel C**	
	(1)	(2)	(1)	(2)	(1)	(2)
	ret	ret	ret	ret	ret	ret
Δna	0.084***	0.083***				
	(12.060)	(11.823)				
$na\ shock^1$			0.085***	0.087***		
			(13.121)	(12.707)		
$na\ shock^2$					0.076***	0.074***
					(13.115)	(12.905)
na		0.001*		-0.001		0.001
		(1.929)		(-1.335)		(1.419)
Cons	-0.003	0.009	0.099***	0.093***	-0.002	0.007
	(-0.349)	(0.846)	(7.165)	(6.998)	(-0.246)	(0.702)
Obs	13,522	13,522	13,404	13,404	12,163	12,163
Adj R^2	0.085	0.088	0.086	0.094	0.087	0.090

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Table 3: **Value Relevance Analysis with Controls**

This table shows the results for the cross-sectional value-relevance tests controlling for additional variables. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ret	ret	ret	ret	ret	ret	ret
Δna	0.084***	0.101***	0.083***	0.083***	0.082***	0.083***	0.081***
	(12.060)	(4.299)	(13.288)	(13.436)	(13.236)	(12.856)	(12.905)
Δvol		-0.005	0.012***	0.011***	0.012***	0.012***	0.012***
		(-0.340)	(4.908)	(4.767)	(5.057)	(4.892)	(4.828)
$\Delta \log(Google+1)$			0.012	0.013	0.003*	0.012	0.003**
			(1.320)	(1.377)	(1.924)	(1.320)	(2.316)
$\Delta current$				0.038			0.053
				(0.319)			(0.419)
$\Delta balance1$					0.286**		0.340**
					(1.991)		(1.978)
$\Delta balance2$						-0.040*	-0.044*
						(-1.826)	(-1.714)
Cons	-0.003	-0.005	-0.001	-0.002	-0.004	-0.001	-0.004
	(-0.349)	(-0.546)	(-0.146)	(-0.211)	(-0.384)	(-0.140)	(-0.477)
Obs	13,522	13,418	11,621	11,621	11,432	11,354	11,181
Adj R^2	0.0847	0.0952	0.121	0.133	0.123	0.147	0.163

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Table 4: **Value Relevance Analysis–Adjusting for Expected Returns with Factor Models**

This table shows the results for the cross-sectional value-relevance tests using returns adjusted for expected returns. The regressions are based on Fama-MacBeth regression method. res^1 is the residual return based on the cryptocurrency CAPM model and res^2 is the residual return based on the cryptocurrency 3-factor model. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	Panel A		**Panel B**	
	(1)	(2)	(1)	(2)
	res^1	res^1	res^2	res^2
Δna	0.087***	0.086***	0.084***	0.083***
	(11.976)	(12.177)	(11.786)	(12.031)
na		0.000		-0.001
		(0.162)		(-0.911)
Cons	-0.008	-0.007	-0.003	-0.010
	(-1.419)	(-0.862)	(-0.796)	(-1.103)
Obs	12,053	12,053	12,053	12,053
Adj R^2	0.077	0.078	0.058	0.057

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Table 5: **Value-Relevance Analysis–Subsample by Persistence and Variability**

This table reports the results of the cross-sectional tests in the subsamples based on the above median and below median persistence or variability of new address growth. Panel A reports results based on the persistence of new address growth and Panel B reports results based on the variability of new address growth. $1_{\{v>Med\}}$ is an indicator variable which equals to one if the persistence or the variability of new address growth is greater than the sample median and zero otherwise. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A: Persistence	(1)	(2)	(3)
	ret	ret	ret
Sample	<Med	>Med	Full
Δna	0.077***	0.105***	0.077***
	(7.935)	(11.327)	(7.935)
$1_{\{p>Med\}}$			-0.003
			(-1.042)
$\Delta na * 1_{\{p>Med\}}$			0.026**
			(2.168)
Constant	-0.004	-0.007	-0.004
	(-0.440)	(-0.745)	(-0.440)
Observations	6,644	6,537	13,181
Adj R^2	0.093	0.118	0.119
Panel B: Variability	(1)	(2)	(3)
	ret	ret	ret
Sample	<Med	>Med	Full
Δna	0.123***	0.073***	0.123***
	(11.511)	(10.026)	(11.511)
$1_{\{v>Med\}}$			0.000
			(0.086)
$\Delta na * 1_{\{v>Med\}}$			-0.044***
			(-4.422)
Constant	-0.005	-0.005	-0.005
	(-0.607)	(-0.572)	(-0.607)
Observations	6,709	6,587	13,296
Adj R^2	0.128	0.080	0.111

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Table 6: **Value Relevance Analysis–Subsample by Network Size**

This table reports the results of the cross-sectional value-relevance tests in the subsamples based on the network size of cryptocurrencies. $1_{\{A>Med\}}$ is an indicator variable which equals to one if the total address of the cryptocurrency is above the sample median in the previous period and zero otherwise. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)
	ret	ret	ret
Sample	<Med	>Med	Full
Δna	0.080***	0.103***	0.080***
	(9.850)	(11.423)	(9.850)
$1_{\{A>Med\}}$			-0.002
			(-0.511)
$\Delta na * 1_{\{A>Med\}}$			0.023**
			(2.136)
Constant	-0.004	-0.006	-0.004
	(-0.450)	(-0.627)	(-0.450)
Observations	6,613	6,906	13,519
Adj R^2	0.085	0.106	0.108

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Table 7: **Slow Reaction to New Address Surprises**

This table shows results of predicting one-, two-, three-periods ahead cryptocurrency returns using new address growths. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1) ret_{+1}	(2) ret_{+2}	(3) ret_{+3}	(4) ret_{+1}	(5) ret_{+2}	(6) ret_{+3}
Δna	-0.005	0.005	-0.005	-0.006	0.000	-0.000
	(-1.117)	(1.194)	(-1.115)	(-1.039)	(0.097)	(-0.012)
beta				0.001	0.007*	0.007*
				(0.158)	(1.732)	(1.850)
size				-0.001*	-0.001*	-0.001
				(-1.743)	(-1.805)	(-1.399)
mom				0.011	0.042***	-0.023
				(0.537)	(2.684)	(-1.358)
Cons	-0.002	-0.002	-0.002	0.022	0.022	0.012
	(-0.248)	(-0.243)	(-0.250)	(1.137)	(1.141)	(0.643)
Observations	13,517	13,408	13,298	12,486	12,377	12,267
Adj R^2	0.010	0.005	0.006	0.059	0.041	0.040

Table 8: **Return and Future New Address Growth**

This table shows the results for predicting next period new address growths using current cryptocurrency returns. Time fixed effect is included and standard errors are double clustered at the coin level and time level. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1) Δna_{+1}	(2) Δna_{+2}	(3) Δna_{+3}
ret	0.055	-0.049	-0.071***
	(1.200)	(-1.191)	(-2.633)
Constant	-0.019***	-0.017***	-0.015***
	(-47.404)	(-43.316)	(-37.540)
Obs	13,519	13,451	13,374
Adj R^2	0.052	0.052	0.053

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Table 9: **Portfolio Sorts Based on Cryptocurrency Price-to-New Address Ratio**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios. Panel A reports the means, standard deviations, and Sharpe ratios of the value-weighted portfolios excess returns. Panel B reports the factor exposures of the portfolios sorted on price-to-new address ratios. The factor models include the cryptocurrency CAPM model and the cryptocurrency 3-factor model. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A					
	Low	2	3	High	Low–High
Mean	0.015*	0.008	0.006	-0.004	0.019***
	(1.682)	(0.769)	(0.550)	(-0.553)	(3.638)
SD	0.114	0.142	0.135	0.105	0.068
SR	0.132	0.056	0.044	-0.038	0.279

Panel B	Alpha		CMKT		CSIZE		CMOM		R^2
Low	0.003	(1.481)	1.025***	(46.655)					0.927
2	-0.003	(-0.382)	1.057***	(17.088)					0.632
3	-0.004	(-0.645)	0.970***	(15.572)					0.588
High	-0.013***	(-2.819)	0.810***	(19.179)					0.684
Low–High	0.016***	(3.387)	0.218***	(4.789)					0.119
Low	0.004	(1.559)	1.029***	(47.160)	0.058*	(1.971)	-0.051**	(-2.439)	0.933
2	-0.007	(-1.254)	1.153***	(22.615)	0.637***	(9.271)	-0.023	(-0.472)	0.767
3	-0.009*	(-1.684)	1.073***	(21.704)	0.681***	(10.215)	-0.027	(-0.559)	0.758
High	-0.015***	(-3.934)	0.865***	(24.291)	0.390***	(8.127)	-0.062*	(-1.802)	0.790
Low–High	0.019***	(4.131)	0.169***	(3.937)	-0.322***	(-5.580)	0.009	(0.215)	0.271

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Table 10: **Strategy Exposures**

This table reports the results of regressing the excess returns of the long-short strategy based on the price-to-new address ratios on different macro variables. Δc is the changes in the logged consumption. Δdc is the changes in the logged durable consumption. Δip is the changes in the logged industrial production. ist is the changes in the logged investment to consumption ratio similar to Papanikolaou (2011) and Kogan and Papanikolaou (2014). The table reports the point estimates, t-statistics, and the R-squareds of the regressions. The data are from the St louis Fed website and are at the monthly frequencies.

	ret	ret	ret	ret
Δc	-0.412			
	(-0.568)			
Δdc		-0.032		
		(-0.087)		
Δip			-0.136	
			(-0.171)	
ist				0.231
				(0.300)
R-squared	0.008	0.000	0.001	0.002

Table 11: **Subsample Results**

This table reports the subsample results of the long-short strategy based on the price-to-new address ratios. For each week, coins are first sorted into the low and high groups based on the address growths (Panel A) or the cost of arbitrage index (Panel B) of the coins. Then, within each group, the coins are further sorted into four portfolios based on their price-to-new address ratios. The table reports the average excess returns and the t-statistics associated with the excess returns.

Panel A: $\Delta Address$					
	1	2	3	4	1–4
Low	1.74	0.97	-0.01	-0.94	2.68***
	(1.62)	(0.84)	(-0.01)	(-1.05)	(3.52)
High	0.18	1.75	0.55	0.00	0.18
	(0.25)	(1.51)	(0.55)	(0.00)	(0.33)
Panel B: Cost of Arbitrage Index					
	1	2	3	4	1–4
Low	1.49*	0.81	0.38	-0.00	1.49***
	(1.69)	(0.80)	(0.36)	(-0.00)	(2.64)
High	2.72**	2.03*	1.66	0.03	2.69***
	(2.24)	(1.73)	(1.58)	(0.03)	(2.71)

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Variable Definitions:

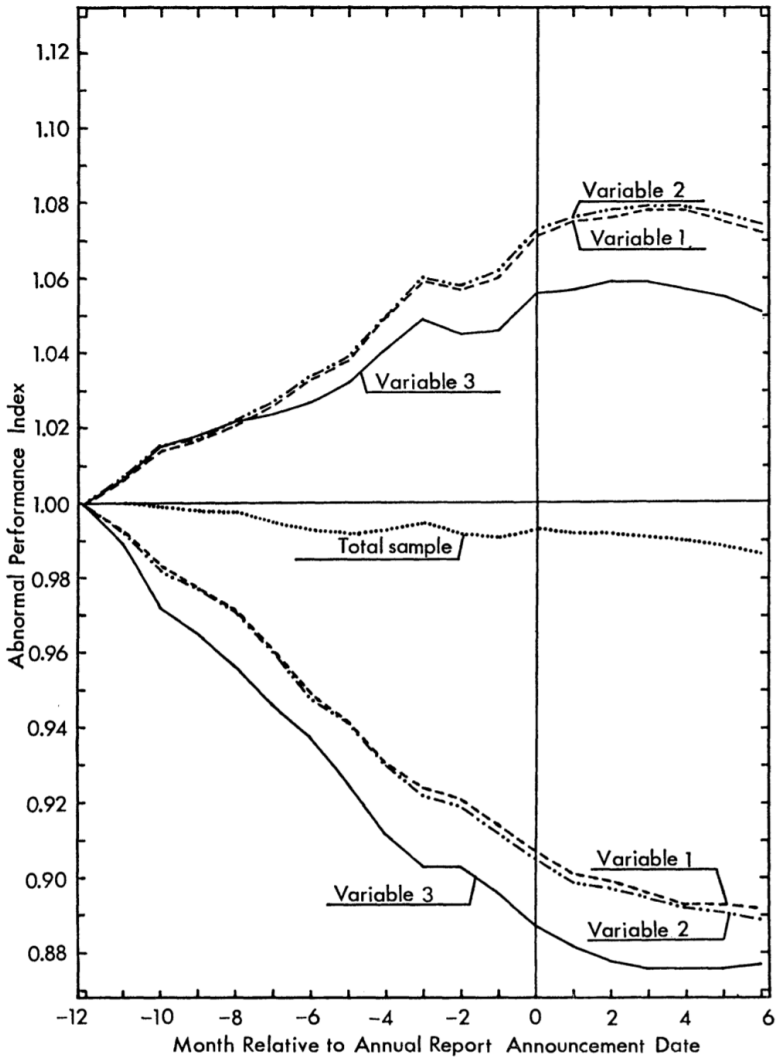
- ret: Logged cryptocurrency return
- na: Logged number of new address per coin
- Δna: Weekly changes in logged number of new address per coin
- pa: Logged price-to-address ratio, or $\log\left(\frac{Price}{NA}\right)$
- vol: Logged transaction volume
- $log\,(Google+1)$: Logged of Google search of the coin plus one
- $current$: Logged total amount of circulating supply
- $balance1$: Logged percentage of supply held by the top 1% addresses
- $balance2$: Logged total amount of coins held on exchange addresses
- $\log(size)$: Logged market capitalization
- $gini$: Gini index of coin holding

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Online Appendix of "Accounting for Cryptocurrency Value"

Figure A.1: **Ball and Brown (1968) Graph**

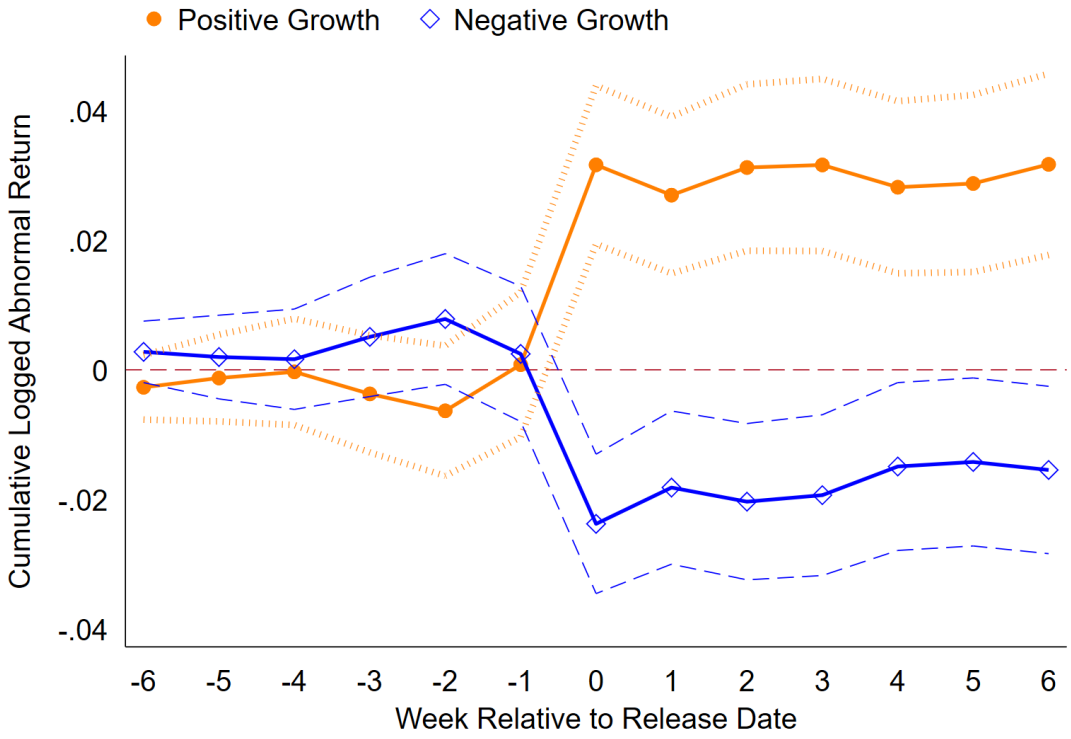
The source of the graph is Ball and Brown (1968).



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Figure A.2: **Cumulative Abnormal Return around New Address Information–Controlling Transfer Volume Growth**

This figure plots the cumulative average logged abnormal return around the disclosure of new address information for positive and negative new address growth subgroups controlling for changes in logged transaction volume. Dashed lines represent the 5-percent confidence intervals.



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Table A.1: **Abnormal Volatility**

This table shows the results for the cross-sectional abnormal volatility tests. Panel A reports results where the abnormal return volatility is calculated using raw returns and Panel B reports results where the abnormal return volatility is calculated using the CAPM model adjusted returns. Abnormal return volatility is calculated as the difference of the logged standard deviation of returns of the week and the logged average standard deviation of returns of the previous four weeks. Time fixed effects are included for all specifications. Standard errors are double clustered at the coin level and time level. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A	(1) $avar^1$	(2) $avar^1$	(3) $avar^1$	(4) $avar^1$	(5) $avar^1$	(6) $avar^1$
$\|\Delta na\|$	0.081***	0.082***	0.085***	0.077***	0.081***	0.081***
	(4.330)	(4.321)	(4.728)	(4.027)	(4.103)	(4.352)
$\|\Delta balance1\|$		0.110				0.085
		(0.678)				(0.536)
$\|\Delta balance2\|$			0.052***			0.050***
			(3.805)			(3.770)
$\|\Delta vol\|$				0.008		0.010
				(0.805)		(1.028)
$\|\Delta current\|$					0.002	-0.032
					(0.018)	(-0.175)
Cons	-0.114***	-0.116***	-0.118***	-0.117***	-0.114***	-0.123***
	(-17.575)	(-17.402)	(-18.713)	(-14.095)	(-17.531)	(-14.027)
Obs	13,137	12,765	12,671	13,115	13,137	12,481
Adj R^2	0.283	0.277	0.286	0.283	0.283	0.283

Panel B	(1) $avar^2$	(2) $avar^2$	(3) $avar^2$	(4) $avar^2$	(5) $avar^2$	(6) $avar^2$
$\|\Delta na\|$	0.124***	0.124***	0.123***	0.116***	0.122***	0.115***
	(4.861)	(4.828)	(4.681)	(4.607)	(4.757)	(4.349)
$\|\Delta balance1\|$		0.203				0.155
		(0.873)				(0.686)
$\|\Delta balance2\|$			0.075***			0.071***
			(2.883)			(2.800)
$\|\Delta trans\|$				0.013		0.013
				(1.284)		(1.268)
$\|\Delta current\|$					0.112	0.036
					(0.446)	(0.133)
Cons	-0.137***	-0.139***	-0.140***	-0.143***	-0.137***	-0.148***
	(-15.773)	(-15.510)	(-15.616)	(-14.442)	(-15.800)	(-14.142)
Obs	11,597	11,255	11,244	11,592	11,597	11,086
Adj R^2	0.146	0.141	0.146	0.147	0.146	0.143

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Table A.2: **Summary Statistics of Other Variables**

This table shows the summary statistics of the other blockchain variables. Panel A reports the mean, standard deviation, 10% percentile, median and 90% percentile values of the variables. Panel B reports the correlation matrix of the variables.

Panel A	Mean	SD	10%	Median	90%
$\Delta bal1$	0.002	0.046	-0.007	0.000	0.011
$\Delta bal2$	0.022	0.591	-0.014	0.000	0.077
Δvol	0.008	1.002	-0.889	-0.000	0.931
$\Delta curr$	0.012	0.105	-0.000	0.000	0.013
nvt	5.050	2.245	2.617	4.916	7.556

Panel B	$\Delta bal1$	$\Delta bal2$	$\Delta trans$	$\Delta curr$	nvt
$\Delta bal1$	1.000				
$\Delta bal2$	0.008	1.000			
Δvol	0.010	0.040	1.000		
$\Delta curr$	0.034	0.043	-0.001	1.000	
nvt	-0.015	-0.035	-0.253	-0.100	1.000

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Table A.3: **Value Relevance Analysis–ERC20**

This table shows the results for the cross-sectional value-relevance tests. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	ret	ret	ret	ret	ret	ret	ret	ret
Δna	0.079***	0.078***	0.079***	0.080***	0.078***	0.079***	0.077***	0.076***
	(12.140)	(12.058)	(12.031)	(14.052)	(12.161)	(11.757)	(13.094)	(12.777)
na		0.001						0.001
		(0.905)						(1.312)
Δvol			0.013***				0.013***	0.012***
			(4.731)				(5.291)	(5.033)
$\Delta current$				0.120			0.002	0.012
				(0.869)			(0.013)	(0.082)
$\Delta balance1$					0.193		0.256	0.298*
					(1.468)		(1.644)	(1.891)
$\Delta balance2$						-0.031	-0.038*	-0.036
						(-1.625)	(-1.680)	(-1.585)
Cons	-0.005	0.004	-0.005	-0.006	-0.005	-0.006	-0.006	0.006
	(-0.543)	(0.294)	(-0.541)	(-0.655)	(-0.552)	(-0.609)	(-0.616)	(0.472)
Obs	11,867	11,867	11,772	11,867	11,851	11,567	11,479	11,479
Adj R^2	0.081	0.091	0.099	0.105	0.094	0.096	0.141	0.148

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Table A.4: **Discount Rate and Value Relevance**

This table reports the results of the cross-sectional tests in the subsamples based on high and low discount rate. $1_{\{d>Med\}}$ is an indicator variable which equals to one if the discount rate is greater than the sample median and zero otherwise. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)
	ret	ret	ret
Sample	<Med	>Med	Full
Δna	0.096***	0.083***	0.096***
	(9.843)	(12.249)	(9.843)
$1_{\{d>Med\}}$			0.002
			(0.662)
$\Delta na * 1_{\{d>Med\}}$			-0.013
			(-1.235)
Constant	-0.004	-0.002	-0.004
	(-0.442)	(-0.195)	(-0.442)
Observations	6,022	5,924	11,946
Adj R^2	0.092	0.098	0.107

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Table A.5: **Distribution of Strategy Returns**

This table shows the average daily long-short strategy returns based on the price-to-new address ratios for each month in the sample. The Recession column shows the NBER recession indicators, where the data are from the St Louis Fed website. "Coin<-20%" is an indicator variable that equals one if the cumulative excess return of the coin market is less than 20% in the month and zero otherwise. "Stock<-5%" is an indicator variable that equals one if the cumulative excess return of the stock market is less than -5% in the month and zero otherwise.

Year-Mon	Strategy (d)	NBER	Coin<-20%	Stock<-5%	Year-Mon	Strategy (d)	NBER	Coin<-20%	Stock<-5%
201802	1.83%	0	0	0	201911	-0.29%	0	0	0
201803	0.12%	0	1	0	201912	0.16%	0	0	0
201804	1.11%	0	0	0	202001	-0.04%	0	0	0
201805	0.43%	0	0	0	202002	-0.59%	0	0	1
201806	0.68%	0	0	0	202003	-0.08%	1	1	1
201807	1.08%	0	0	0	202004	0.59%	1	0	0
201808	0.47%	0	0	0	202005	0.08%	0	0	0
201809	-0.08%	0	0	0	202006	-0.41%	0	0	0
201810	-0.39%	0	0	1	202007	0.51%	0	0	0
201811	-0.14%	0	1	0	202008	-0.18%	0	0	0
201812	0.18%	0	0	1	202009	0.30%	0	0	0
201901	-0.14%	0	0	0	202010	1.47%	0	0	0
201902	-0.37%	0	0	0	202011	1.15%	0	0	0
201903	0.33%	0	0	0	202012	0.40%	0	0	0
201904	0.72%	0	0	0	202101	-0.07%	0	0	0
201905	0.58%	0	0	1	202102	-0.07%	0	0	0
201906	0.16%	0	0	0	202103	0.74%	0	0	0
201907	0.58%	0	0	0	202104	1.00%	0	0	0
201908	0.46%	0	0	0	202105	-0.13%	0	0	0
201909	-0.11%	0	0	0	202106	0.41%	0	0	0
201910	-0.11%	0	0	0					

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Table A.6: **Cryptocurrency Price-to-New Address Effect–Different Horizon**

This table shows portfolio sorting results over different horizons based on cryptocurrency price-to-new address ratios. The Low–High row reports the average long-short strategy returns based on cryptocurrency price-to-new address ratios. The C1 Alpha and C3 Alpha rows report the cryptocurrency CAPM model and the cryptocurrency 3-factor model adjusted alphas, respectively. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Weeks	+1	+4	+8	+12	+16	+20	+24	+28
Low	0.015*	0.017**	0.018**	0.014	0.016*	0.019**	0.017*	0.017*
	(1.682)	(1.990)	(2.033)	(1.541)	(1.787)	(2.106)	(1.887)	(1.891)
2	0.008	0.009	0.005	0.008	0.014	0.016	0.005	0.017
	(0.769)	(0.853)	(0.509)	(0.787)	(1.363)	(1.379)	(0.447)	(1.644)
3	0.006	0.006	0.015	0.011	0.011	0.011	0.019*	0.012
	(0.550)	(0.616)	(1.380)	(1.018)	(0.984)	(1.032)	(1.691)	(1.102)
High	-0.004	0.000	0.005	0.001	0.008	0.009	0.008	0.012
	(-0.553)	(0.030)	(0.627)	(0.112)	(1.004)	(1.083)	(0.928)	(1.469)
Low–High	0.019***	0.018***	0.013**	0.012**	0.009	0.010*	0.010	0.005
	(3.638)	(3.170)	(2.322)	(2.214)	(1.502)	(1.660)	(1.585)	(0.843)
C1 Alpha	0.016***	0.015***	0.009*	0.009*	0.004	0.005	0.005	-0.000
	(3.387)	(2.819)	(1.758)	(1.785)	(0.797)	(0.928)	(0.941)	(-0.050)
C3 Alpha	0.019***	0.017***	0.010**	0.010**	0.006	0.008*	0.009*	0.004
	(4.131)	(3.458)	(2.131)	(2.280)	(1.420)	(1.816)	(1.830)	(0.772)

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Table A.7: **Fama-MacBeth Regression Results**

This table shows the results for predicting next period returns using the price-to-new address ratios. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)
	RET_{+1}	RET_{+1}	RET_{+1}	RET_{+1}	RET_{+1}
pa	-0.004***	-0.003***	-0.003***	-0.003**	-0.003**
	(-2.805)	(-2.696)	(-2.674)	(-2.563)	(-2.554)
beta		0.004			0.003
		(0.877)			(0.755)
size			-0.002***		-0.002**
			(-2.679)		(-2.065)
mom				0.014	0.019
				(0.862)	(1.138)
Constant	0.054**	0.051**	0.089***	0.045**	0.073**
	(2.539)	(2.468)	(3.015)	(2.142)	(2.530)
Observations	13,455	12,449	13,455	13,455	12,449
R-squared	0.017	0.049	0.031	0.058	0.102

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Table A.8: **Portfolio Sorts–Tercile and Quintile**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios using tercile and quintile portfolios. Mean is the average weekly excess returns of the portfolios and the long-short strategy. SD is the standard deviation of the weekly excess returns of the portfolios and the long-short strategy. SR is the weekly Sharpe ratio of the excess returns of the portfolios and the long-short strategy. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A: Tercile				
	Low	2	High	Low–High
Mean	0.013	0.012	-0.003	0.016***
	(1.552)	(1.111)	(-0.316)	(3.122)
SD	0.113	0.137	0.106	0.066
SR	0.115	0.088	-0.028	0.242

Panel B: Quintile						
	Low	2	3	4	High	Low–High
Mean	0.012	0.008	0.016	0.005	-0.004	0.016***
	(1.422)	(0.868)	(1.464)	(0.456)	(-0.451)	(2.841)
SD	0.114	0.128	0.143	0.134	0.107	0.073
SR	0.105	0.063	0.112	0.037	-0.037	0.219

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Table A.9: **Portfolio Sorts–Top 30 Cryptocurrencies**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios using the largest 30 cryptocurrencies by market capitalization. Mean is the average weekly excess returns of the portfolios and the long-short strategy. SD is the standard deviation of the weekly excess returns of the portfolios and the long-short strategy. SR is the weekly Sharpe ratio of the excess returns of the long-short strategy. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A				
	Low	2	High	Low–High
Mean	0.012	0.006	-0.003	0.015***
	(1.400)	(0.544)	(-0.402)	(2.814)
SD	0.111	0.141	0.107	0.069
SR	0.108	0.043	-0.028	0.217

Panel B	Alpha		CMKT		CSIZE		CMOM		R^2
Low	0.001	(0.579)	1.028***	(86.532)					0.978
2	-0.005	(-0.664)	1.014***	(15.720)					0.592
High	-0.012**	(-2.418)	0.815***	(18.082)					0.658
Low–High	0.013**	(2.520)	0.215***	(4.583)					0.110
Low	0.001	(0.654)	1.027***	(83.809)	-0.000	(-0.017)	-0.010	(-0.814)	0.978
2	-0.009	(-1.639)	1.117***	(21.231)	0.678***	(9.565)	-0.022	(-0.444)	0.747
High	-0.014***	(-3.355)	0.873***	(22.036)	0.394***	(7.382)	-0.038	(-0.987)	0.754
Low–High	0.015***	(3.382)	0.157***	(3.721)	-0.387***	(-6.790)	0.027	(0.661)	0.326

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Table A.10: **Portfolio Sorts–Exclude Bitcoin**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios excluding Bitcoin. Mean is the average weekly excess returns of the portfolios and the long-short strategy. SD is the standard deviation of the weekly excess returns of the portfolios and the long-short strategy. SR is the weekly Sharpe ratio of the excess returns of the portfolios and the long-short strategy. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A					
	Low	2	3	High	Low–High
Mean	0.009	0.013	0.008	-0.004	0.013**
	(0.886)	(1.170)	(0.726)	(-0.537)	(2.424)
SD	0.133	0.147	0.138	0.105	0.072
SR	0.068	0.088	0.058	-0.038	0.181

Panel B	Alpha		CMKT		CSIZE		CMOM		R^2
Low	0.004	(1.181)	1.096***	(42.285)					0.913
2	0.008	(1.309)	1.076***	(20.984)					0.721
3	0.003	(0.469)	1.002***	(20.480)					0.712
High	-0.008*	(-1.958)	0.771***	(21.554)					0.732
Low–High	0.012**	(2.494)	0.325***	(8.057)					0.276
Low	0.006**	(2.002)	1.086***	(45.745)	-0.054	(-1.548)	-0.151***	(-5.975)	0.928
2	0.006	(1.024)	1.066***	(23.157)	0.430***	(6.374)	-0.001	(-0.019)	0.779
3	0.001	(0.230)	0.989***	(22.764)	0.400***	(6.286)	-0.056	(-1.218)	0.776
High	-0.007*	(-1.859)	0.760***	(22.322)	0.110**	(2.216)	-0.116***	(-3.214)	0.761
Low–High	0.013***	(2.786)	0.326***	(8.202)	-0.164***	(-2.820)	-0.035	(-0.816)	0.309

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Table A.11: **Portfolio Sorts Based on Cryptocurrency Price-to-New Address Ratio–Equal Weighted**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios with equal-weighting. Panel A reports the means, standard deviations, and Sharpe ratios of the equal-weighted portfolios excess returns. Panel B reports the factor exposures of the portfolios sorted on price-to-new address ratios. The factor models include the cryptocurrency CAPM model and the cryptocurrency 3-factor model. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A					
	Low	2	3	High	Low–High
Mean	0.016	0.013	0.013	-0.001	0.016***
	(1.601)	(1.218)	(1.329)	(-0.118)	(3.636)
SD	0.129	0.138	0.125	0.108	0.059
SR	0.124	0.094	0.104	-0.009	0.271

Panel B	Alpha		CMKT		CSIZE		CMOM		R^2
Low	0.006	(0.907)	0.907***	(14.893)					0.565
2	0.002	(0.372)	1.002***	(16.194)					0.607
3	0.003	(0.543)	0.919***	(16.487)					0.615
High	-0.009*	(-1.936)	0.817***	(18.069)					0.658
Low–High	0.016***	(3.449)	0.092**	(2.194)					0.028
Low	-0.001	(-0.145)	1.028***	(28.373)	0.842***	(17.384)	-0.024	(-0.678)	0.856
2	-0.004	(-1.261)	1.146***	(37.961)	0.939***	(23.089)	-0.011	(-0.366)	0.913
3	-0.002	(-0.767)	1.045***	(35.447)	0.823***	(20.721)	-0.013	(-0.471)	0.900
High	-0.012***	(-4.105)	0.898***	(31.431)	0.572***	(14.843)	-0.088***	(-3.194)	0.872
Low–High	0.013***	(3.018)	0.143***	(3.581)	0.301***	(5.596)	0.060	(1.554)	0.180

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Table A.12: **Portfolio Sorts–First Half and Second Half**

This table shows portfolio sorting results based on cryptocurrency price-to-new address ratios for the first half and the second half of the sample. Mean is the average weekly excess returns of the portfolios and the long-short strategy. SD is the standard deviation of the weekly excess returns of the portfolios and the long-short strategy. SR is the weekly Sharpe ratio of the excess returns of the portfolios and the long-short strategy. Variable definition is documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

Panel A: First Half					
	Low	2	3	High	Low–High
Mean	-0.002	-0.020	-0.016	-0.021*	0.018**
	(-0.212)	(-1.589)	(-1.291)	(-1.965)	(2.619)
SD	0.104	0.123	0.121	0.107	0.070
SR	-0.019	-0.163	-0.132	-0.196	0.257
Panel B: Second Half					
	Low	2	3	High	Low–High
Mean	0.036**	0.045**	0.034*	0.017	0.019**
	(2.572)	(2.456)	(1.978)	(1.531)	(2.525)
SD	0.123	0.157	0.148	0.098	0.065
SR	0.293	0.287	0.230	0.173	0.292

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Table A.13: **Equity Factors Exposures**

This table shows exposures to the equity factor models for the portfolios based on cryptocurrency price-to-new address ratios. The equity factor models are the CAPM, Fama-French 3-factor, and the Fama-French 5-factor models. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	Alpha		MKTRF		SMB		HML		RMW		CMA		R^2
Low	0.012	(1.388)	0.855**	(2.563)									0.037
2	0.004	(0.360)	1.480***	(3.614)									0.071
3	0.001	(0.146)	1.373***	(3.515)									0.068
High	-0.007	(-0.940)	0.963***	(3.168)									0.056
Low–High	0.019***	(3.669)	-0.110	(-0.543)									0.002
Low	0.012	(1.450)	0.749**	(2.177)	0.652	(1.098)	0.234	(0.586)					0.051
2	0.003	(0.313)	1.352***	(3.192)	0.891	(1.219)	-0.282	(-0.573)					0.080
3	0.002	(0.206)	1.376***	(3.395)	-0.079	(-0.113)	0.297	(0.630)					0.070
High	-0.007	(-0.839)	0.918***	(2.919)	0.234	(0.431)	0.342	(0.936)					0.065
Low–High	0.019***	(3.611)	-0.170	(-0.815)	0.419	(1.162)	-0.110	(-0.455)					0.010
Low	0.014	(1.566)	0.629*	(1.689)	0.393	(0.575)	0.679	(1.148)	-0.550	(-0.552)	-1.084	(-0.864)	0.057
2	0.004	(0.393)	1.352***	(2.946)	0.596	(0.707)	0.001	(0.001)	-0.904	(-0.732)	-0.001	(-0.001)	0.083
3	0.003	(0.269)	1.238***	(2.820)	-0.159	(-0.198)	0.600	(0.859)	0.030	(0.025)	-1.241	(-0.838)	0.074
High	-0.005	(-0.653)	0.839**	(2.471)	-0.146	(-0.234)	0.835	(1.546)	-1.004	(-1.099)	-0.713	(-0.622)	0.074
Low–High	0.019***	(3.515)	-0.213	(-0.943)	0.547	(1.318)	-0.163	(-0.454)	0.479	(0.788)	-0.384	(-0.504)	0.015

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Table A.14: **Cryptocurrency Market Factor Excluding Bitcoin**

This table shows portfolio sorting results controlling for the cryptocurrency CAPM model and the cryptocurrency 3-factor model, where the cryptocurrency market factor is constructed as the value-weighted cryptocurrency returns excluding bitcoin over the risk-free rate (CMKT2). Mean is the average weekly excess returns of the portfolios and the long-short strategy. SD is the standard deviation of the weekly excess returns of the portfolios and the long-short strategy. SR is the weekly Sharpe ratio of the excess returns of the portfolios and t0he long-short strategy. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	Alpha		CMKT2		CSIZE		CMOM		R^2
Low	0.009**	(2.211)	0.850***	(23.482)					0.763
2	0.003	(0.534)	1.005***	(18.769)					0.674
3	0.001	(0.157)	0.990***	(20.944)					0.721
High	-0.008**	(-1.994)	0.771***	(21.613)					0.733
Low–High	0.018***	(3.581)	0.090**	(2.045)					0.024
Low	0.012***	(3.176)	0.851***	(25.642)	-0.278***	(-5.746)	-0.120***	(-3.388)	0.805
2	0.003	(0.510)	0.992***	(19.480)	0.265***	(3.549)	-0.099*	(-1.828)	0.711
3	0.000	(0.026)	0.976***	(22.863)	0.334***	(5.341)	-0.092**	(-2.031)	0.776
High	-0.008*	(-1.895)	0.760***	(22.398)	0.110**	(2.220)	-0.117***	(-3.243)	0.763
Low–High	0.020***	(4.422)	0.098**	(2.485)	-0.376***	(-6.515)	-0.006	(-0.135)	0.231

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Table A.15: **Fama-MacBeth Regression Results–New Address Growth**

This table shows the results for predicting next period new address growths using the price-to-new address ratios. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1) Δna_{+1}	(2) Δna_{+1}	(3) Δna_{+1}	(4) Δna_{+1}	(5) Δna_{+1}
pa	0.090***	0.078***	0.091***	0.088***	0.078***
	(16.205)	(13.516)	(15.988)	(15.951)	(13.081)
beta		0.006			0.007
		(0.588)			(0.732)
size			0.002		0.002
			(1.031)		(0.812)
mom				0.063	0.043
				(1.554)	(0.950)
Constant	-1.116***	-0.973***	-1.173***	-1.081***	-1.005***
	(-16.455)	(-14.054)	(-12.322)	(-15.969)	(-10.341)
Obs	13,343	12,337	13,343	13,343	12,337
R-squared	0.074	0.080	0.087	0.099	0.117

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Table A.16: **Subsample with Controls**

This table reports the results of the cross-sectional tests in the subsamples based on the above median and below median persistence, variability, discount rate, and size with controls, respectively. $1_{\{X>Med\}}$ is an indicator variable which equals to one if the $X \in \{Persistence, Variability, Discount, Size\}$ is greater than the sample median and zero otherwise. The regressions are based on Fama-MacBeth regression method. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
	ret	ret	ret	ret
X	Persistence	Variability	Discount	Size
Δna	0.075***	0.106***	0.090***	0.075***
	(11.250)	(11.206)	(10.525)	(11.250)
$1_{\{X>Med\}}$	-0.005*	-0.000	0.001	-0.002
	(-1.676)	(-0.011)	(0.474)	(-0.586)
$\Delta na * 1_{\{X>Med\}}$	0.025**	-0.039***	-0.013	0.022**
	(2.458)	(-4.199)	(-1.503)	(2.441)
Δvol	0.007***	0.007***	0.007***	0.008***
	(3.353)	(3.330)	(3.245)	(3.762)
$\Delta current$	-0.043	-0.007	-0.089	0.032
	(-0.255)	(-0.053)	(-0.709)	(0.230)
$\Delta balance1$	0.359**	0.316**	0.298	0.267*
	(2.168)	(2.022)	(1.508)	(1.742)
$\Delta balance2$	-0.038*	-0.035	-0.017	-0.036*
	(-1.800)	(-1.560)	(-0.708)	(-1.700)
Constant	-0.002	-0.006	-0.003	-0.005
	(-0.217)	(-0.624)	(-0.347)	(-0.499)
Observations	12,553	12,655	11,444	12,758
Adj R^2	0.161	0.154	0.150	0.166

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Table A.17: **Value Relevance Analysis–Growth**

This table shows the results for the cross-sectional value-relevance tests. The regressions are based on Fama-MacBeth regression method. The variables are in growth rates instead of logged growth rates. Variable definitions are documented in the Appendix. *, **, and *** represent statistical significance levels at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Ret	*Ret*	*Ret*	*Ret*	*Ret*	*Ret*
gNA	0.079***	0.098***	0.081***	0.075***	0.088***	0.074***
	(13.010)	(3.424)	(11.565)	(14.372)	(6.302)	(13.551)
$gVol$		-0.012				0.007***
		(-0.583)				(3.852)
$gCurrent$			-0.780			0.070
			(-0.807)			(0.629)
$gBalance1$				0.098		0.186
				(0.711)		(1.226)
$gBalance2$					-0.007	-0.016
					(-0.326)	(-0.714)
Cons	0.006	0.003	0.006	0.005	0.007	0.004
	(0.631)	(0.386)	(0.658)	(0.527)	(0.731)	(0.491)
Obs	13,522	13,418	13,522	13,139	13,027	12,759
Adj R^2	0.116	0.127	0.123	0.122	0.137	0.162

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